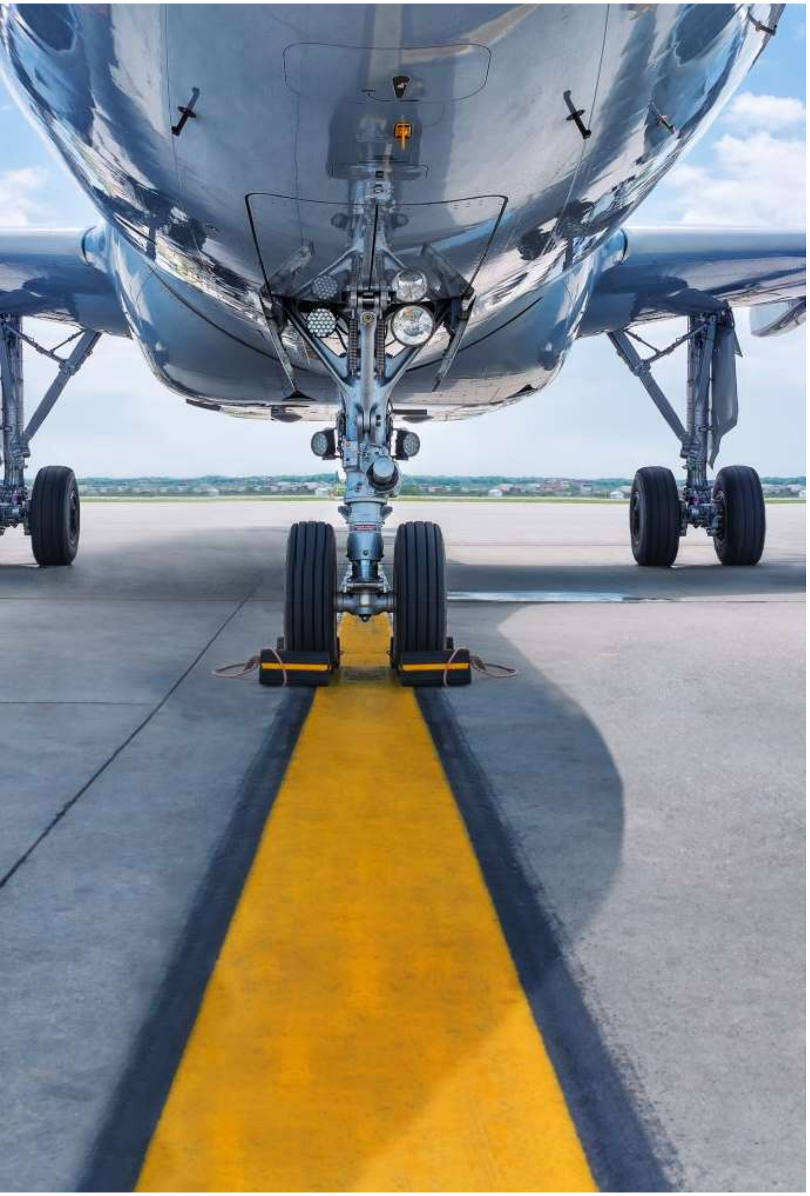
Aircraft Engine Combustion Systems

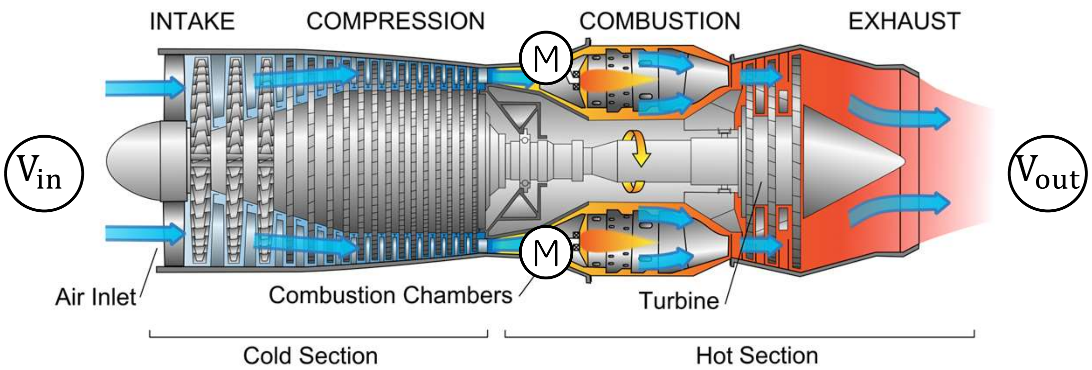


Introduction

With several others, I was sent on loan from the Royal Aircraft Establishment to lead a group on combustion at Power Jets [in 1940]. I had done my thesis on laminar and turbulent diffusion flames and knew the importance of aerodynamics in the combustion process. It surprised me that others did not see that as much care was required in characterizing the aerodynamic features of a combustion chamber as in the design of a blade for a compressor or turbine.

- Sir William Hawthorne, "The Early History of the Aircraft Gas Turbine in Britain,"

Aircraft Engine - Thrust

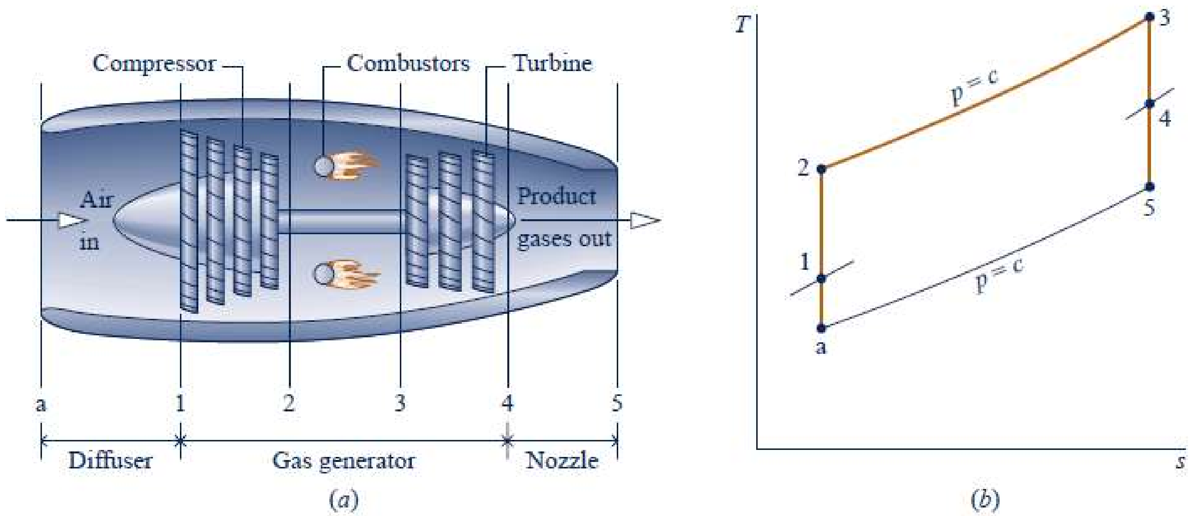


$$\text{FORCE} = \text{MASS} \times \text{ACCELERATION}$$

$$\text{FORCE} = \dot{\text{M}} \times (\text{V}_{\text{out}} - \text{V}_{\text{in}}) = \text{THRUST}$$

Aircraft Engine – Working Cycle

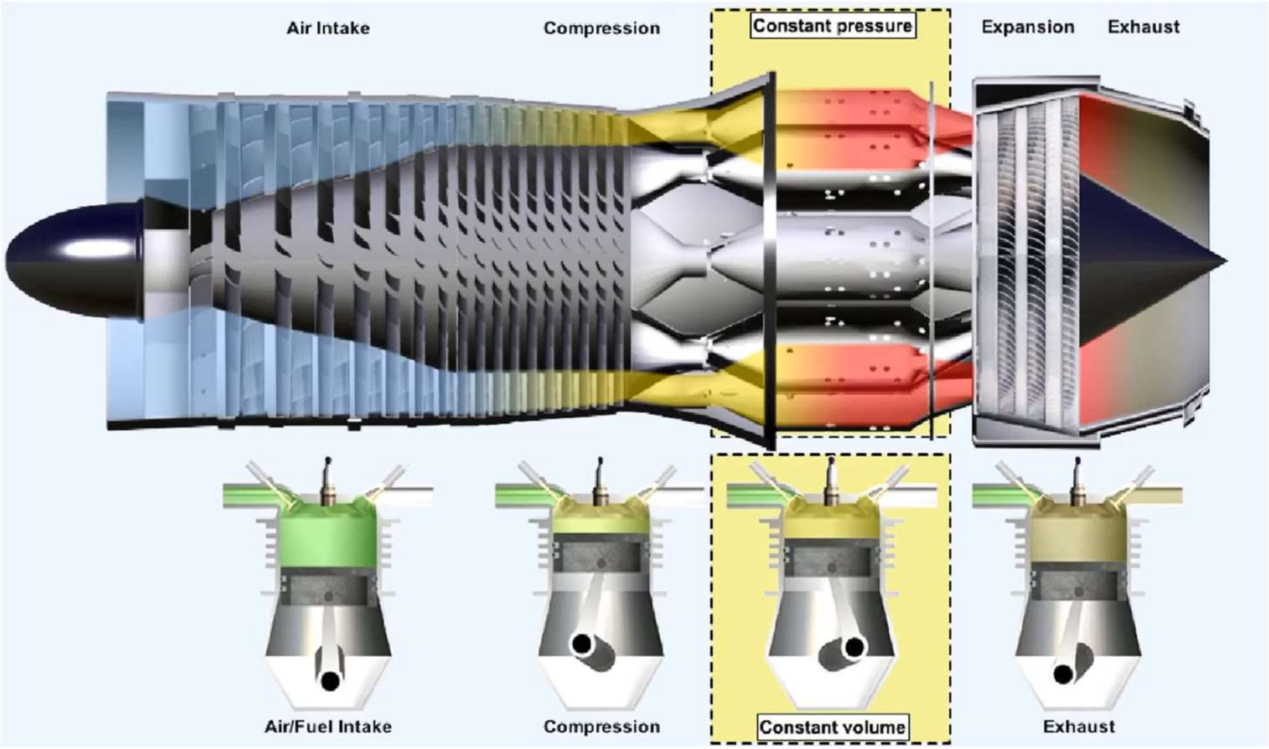
Heat input to the gas turbine Brayton cycle is provided by a combustor. The combustor accepts air from the compressor and delivers it at an elevated temperature to the turbine (ideally with no pressure loss). Combustion are then mixed with the remaining air to arrive at a suitable turbine inlet temperature.



Aircraft Engine – Working Cycle

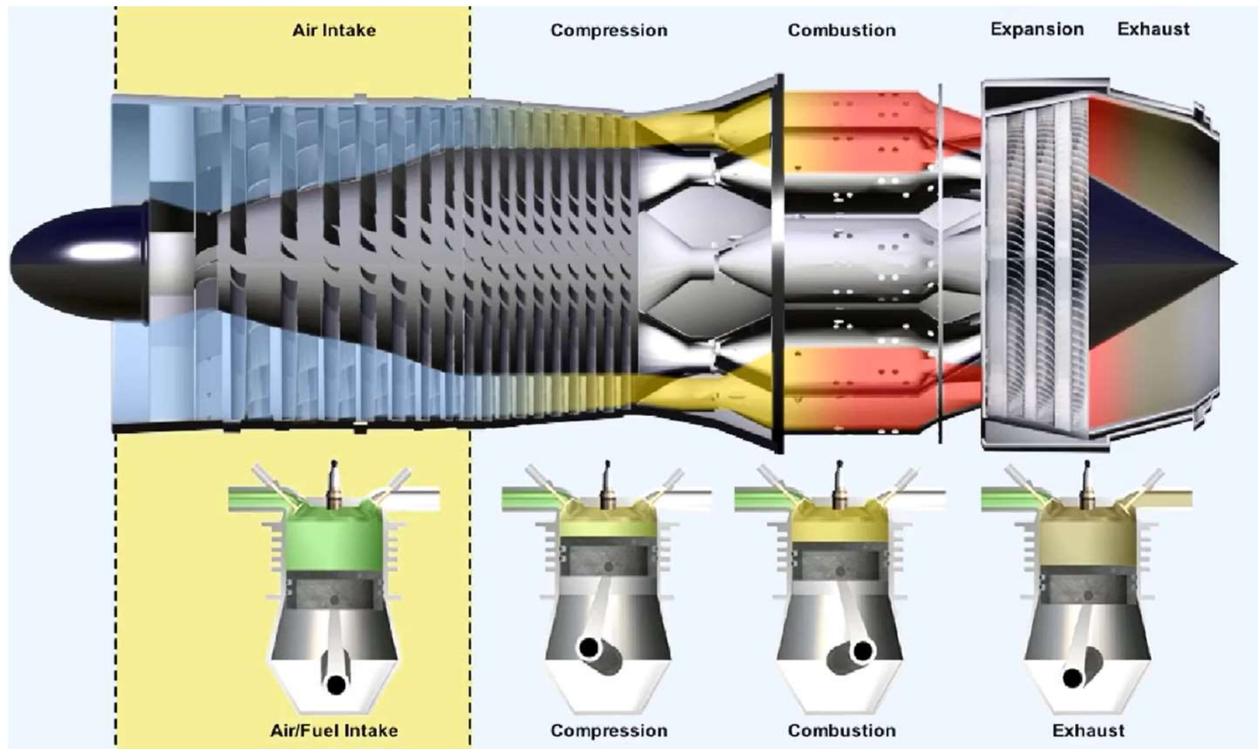
Brayton cycle and Otto Cycle are very similar. Where all the stages are matched between both with one major difference.

In the gas turbine, combustion occurs at constant pressure whereas in the piston engine occurs at a constant volume.



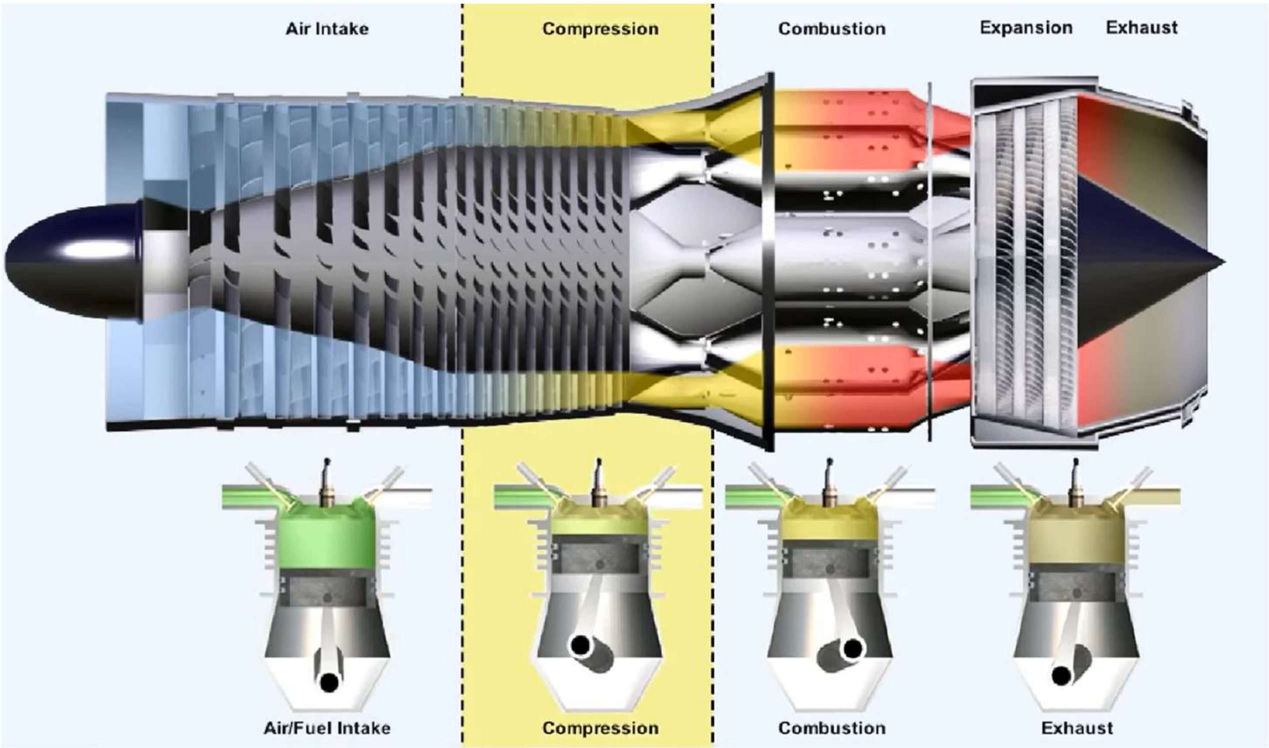
Aircraft Engine – Working Cycle

Air admission



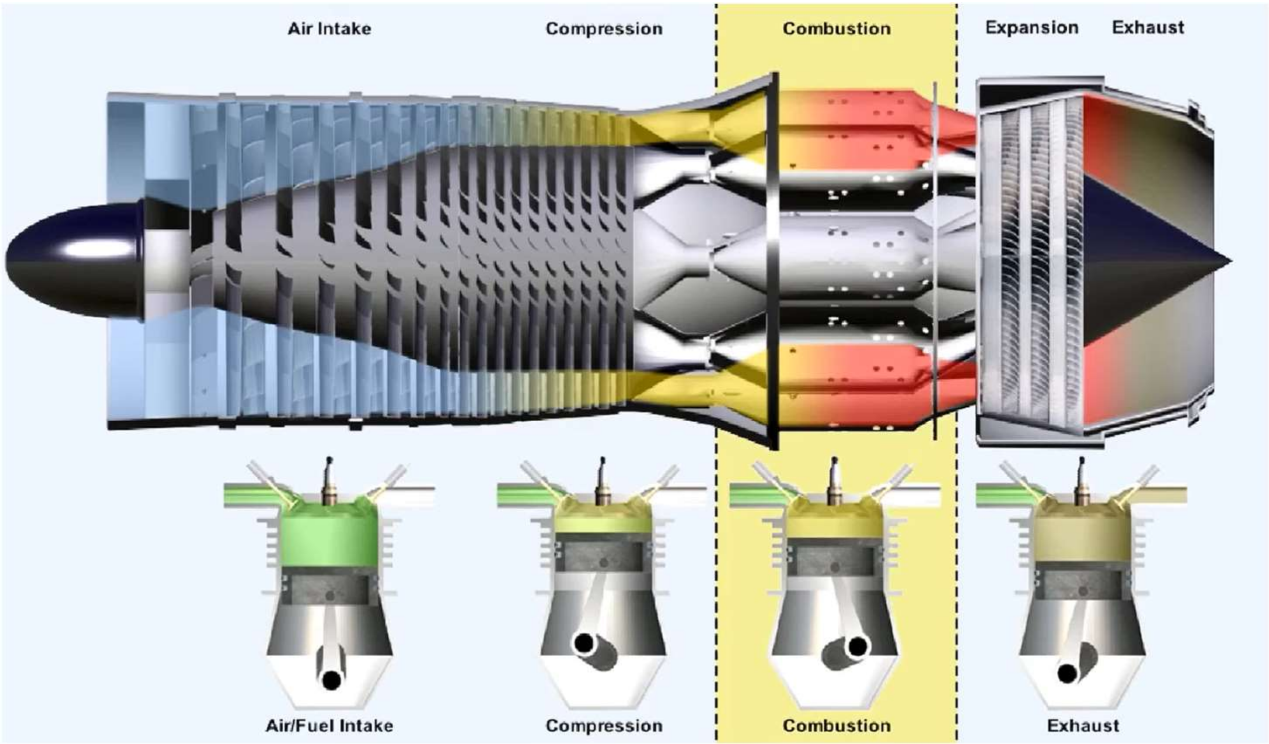
Aircraft Engine – Working Cycle

Air Compression



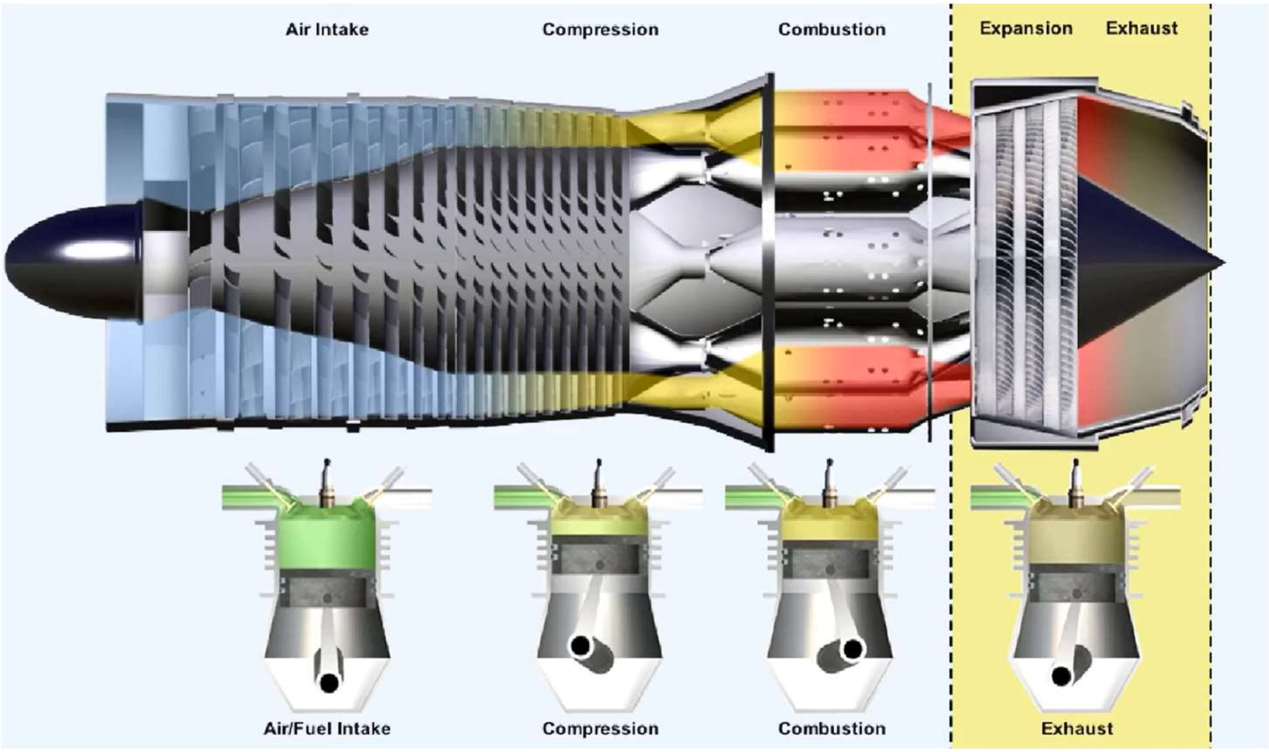
Aircraft Engine – Working Cycle

Combustion



Aircraft Engine – Working Cycle

Expansion and gas release

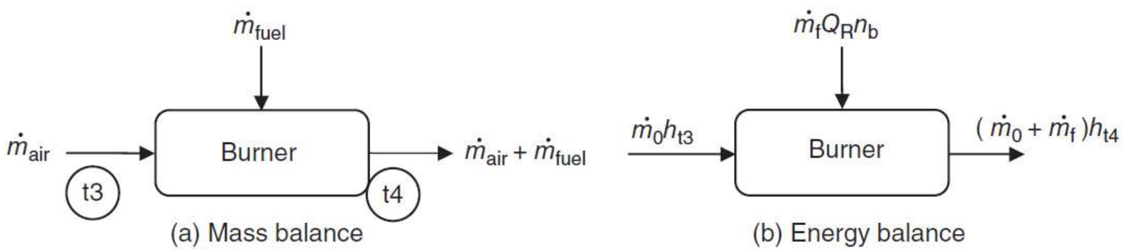


Conceptual Differences

The design of the main burner and afterburner of an airbreathing engine differs in many ways from that of stationary combustion devices. Space (especially length) is at a premium in aircraft applications. The combustion intensity (rate of thermal energy released per unit volume) is very much greater for the main burner of a turbojet (40,000 Btu/s-ft 3) than, for example, the furnace of a typical steam power plant (10 Btu/s-ft3).

Heat and Mass Balance (Reminder)

Combustion temperature depends on the engine pressure ratio, whilst the combustion efficiency depends on the air mass and fuel flow and fuel composition.



$$\eta_{burner} = \frac{\dot{Q}_{useful}}{\dot{Q}_{available}} = \frac{(\dot{m}_a + \dot{m}_f)h_{prod} - \dot{m}_a h_{react}}{\dot{m}_f (LHV)}$$

Example

A gas turbine combustor has an air inlet condition of T_{T_3} = 800 K, P_{T_3} = 2 MPa , and a mass flow rate of $\dot{m}_{air}$ = 50 kg/s, $C_{P_{air}}$ =1.004 kJ/kg·K. A hydrocarbon fuel with ideal heating value LHV = 42,000 kJ/kg is injected in the combustor at a rate of $\dot{m}_{fuel}$ = 1 kg/s. The burner efficiency is η_{burner} = 99.5% and the total pressure at the combustor exit is 96% of the inlet total pressure, i.e., combustion causes a 4% loss in total pressure. The gas properties at the combustor exit $C_{P_{prod}}$ = 1.156 kJ/kg·K. Calculate

a) fuel-to-air ratio f
b) combustor exit temperature Tt4 in K and pt4 in MPa

Solution

The air and the fuel Flow rates are specified at 50 and 1 kg/s, respectively, in the problem, therefore, $f = 1/50 = 0.02$

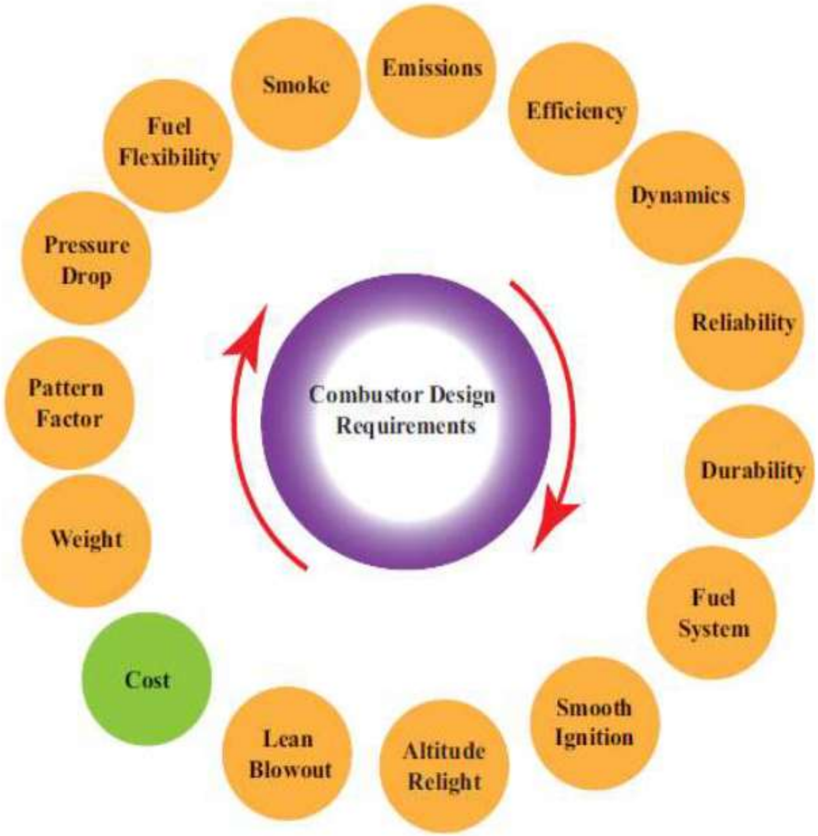
We calculate combustor exit temperature by energy balance

$$\dot{m}_{air} h_{T_{T_3}} + \dot{m}_{fuel} LHV \eta_b = (\dot{m}_{air} + \dot{m}_{fuel}) h_{T_{products}}$$

Therefore

$$T_{prod} = \frac{\left(\frac{c_{P_{air}}}{c_{P_{prod}}}\right) T_{air} + f \cdot LHV \cdot \eta_b \left(\frac{1}{c_{P_{prod}}}\right)}{1+f} = 1390.02\ K$$

And Pprod = 0.96Pair = 1.92 MPa



Design Criteria

WHAT ARE THE MAIN DRIVERS OF THE DESIGN?

Basic Design Criteria

Problem definition:

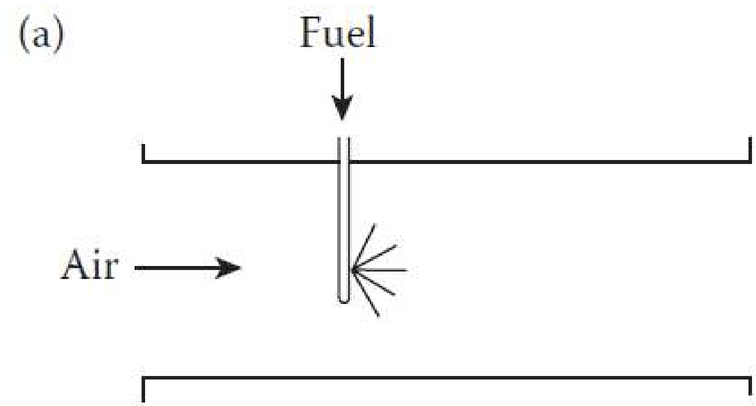
- Enable more efficient use of fuel with less pollution of the atmosphere.
- Combustor must be able to deliver stable and efficient combustion over a wide range of engine operating conditions.
  - Reliable and smooth ignition
  - Wide stability limits (the flame should stay alight)
  - Compressor will be delivering high pressure and temperature gases (~ 550°C at full power settings).
- Temperature limits of the gas exiting from the combustion chamber are imposed by the materials of the turbine (usually not above 1700° C at full power settings).

Basic Design Criteria

Simple desing

Challenge?

- Unacceptable pressure loss due to increased air speed



Design Criteria: Combustor Pressure Loss

Cold Pressure Loss:

- Combustor diffuser (flow Mach number reduction M ~ 0.15 – 0.2)
- Liner losses – Air mixing in secondary and dilution zones

Combustion Loss:

- Proportional to the change in momentum of the flow ($\propto V^2$) due to the density reduction consequence of the heat addition.

Hence:

Combustor pressure loss is given by:

$$\Delta P_{fund} = \tfrac{1}{2}\rho V^2 \times [T_4/T_3 - 1]$$

Design Criteria: Combustor Pressure Loss

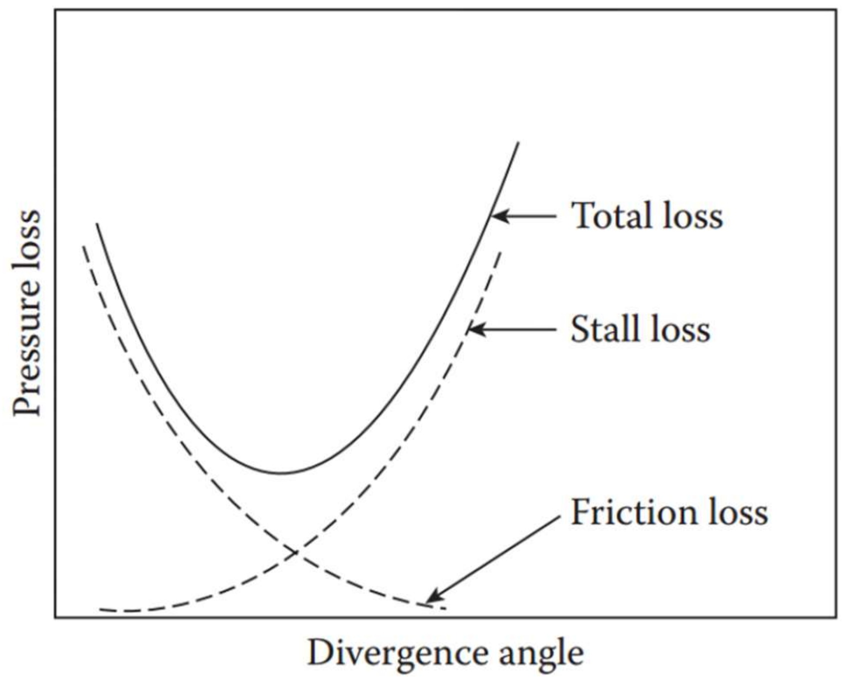
Velocity diffusion prior to both the main burner and afterburner is necessary to reduce total pressure loss due to combustion and to lower velocities significantly to facilitate ignition and flame stabilization. For example, air exiting modern compressors may reach a velocity of 170 m/s. This velocity combined with a main burner temperature ratio of ~ 2.5 would result in a 25% loss in total pressure.

Therefore:

- The design of a combustion chamber should be such that the input speed to the combustor is equal to the output.

Design Criteria: Diffuser performance

The choices of diffuser design are flat wall only, dump only or combined flat wall and dump, where the flat-wall has good diffusion effectiveness, but is long; the dump is short, but has low effectiveness. The combined flat-wall and dump offers an attractive compromise between effectiveness and length for area ratios AR less than 4 and is necessary for area ratios above 4.



Design Criteria: Diffuser performance

The ratio of actual-to-ideal pressure recovery is a logical measure of diffuser performance and is called the diffuser effectiveness or diffusion efficiency

$$\eta_D = \frac{C_P}{C_{P_{id}}}$$

This along with the total pressure loss coefficient is $(\Delta P_t/q_1)_D$ can be then related to the total pressure ratio πD

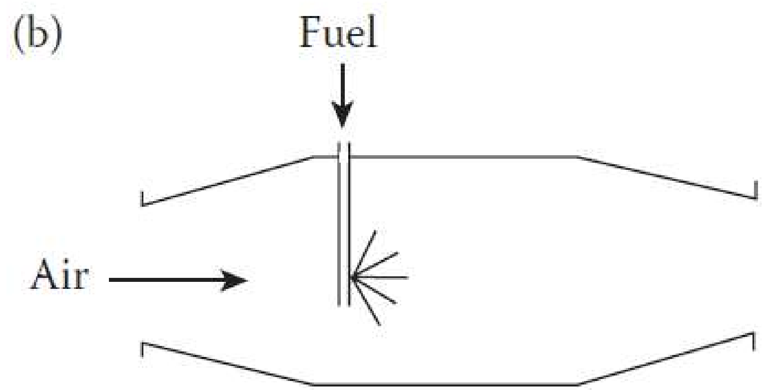
$$\pi_D = \frac{P_{t2}}{P_{t1}} = 1 - \frac{(\Delta P_t/q_1)_D}{1 + 2/\gamma M_1^2} = 1 - \frac{(1 - 1/AR^2)(1 - \eta_D)}{1 + 2/\gamma M_1^2}$$

Basic Design Criteria

Reduced pressure los

Challenges?

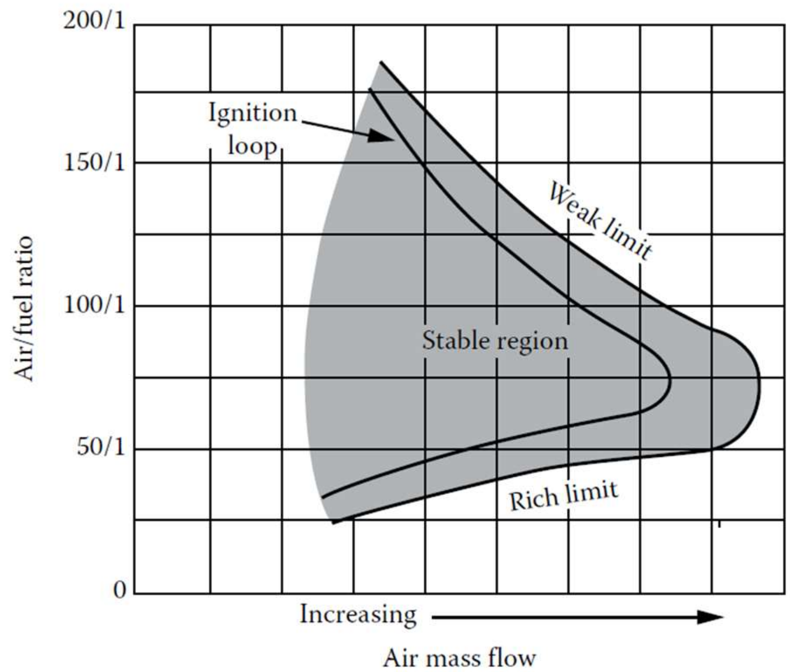
- Combustion stability



Flame characteristics

The flame of the gas turbine is burned at relatively low speeds in contrast to the speed of the air entering the combustor, in order to ensure stable combustion.

The goal is to optimize the air / fuel to get the most power with the least amount of fuel

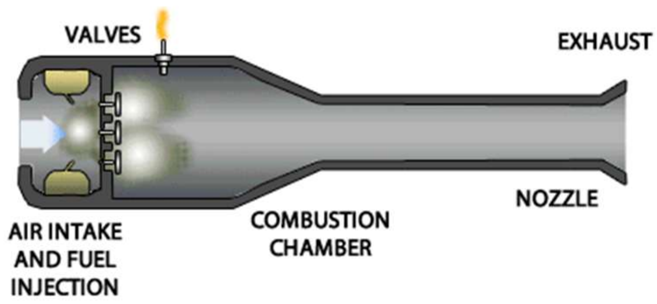


Flame characteristics

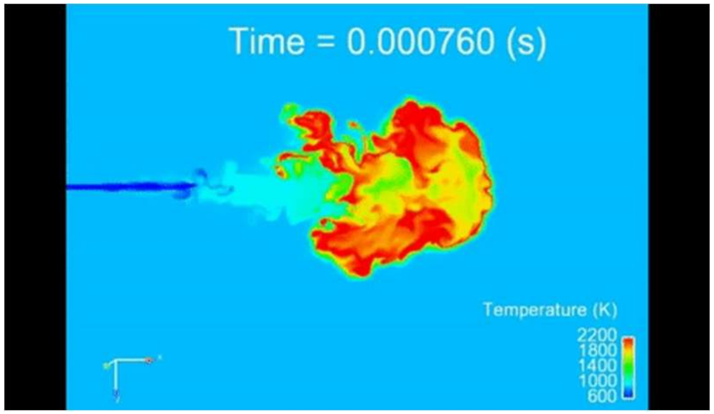
Gas turbine according to its design can accommodate two types of flames, turbulent and laminar and ignition detonation or deflagration.

Both gas turbines and aircraft engines choose to operate with deflagrated premixed flames in an environment that can be laminar or turbulent.

Detonation



Deflagration



Flame Characteristics: Influence on the design

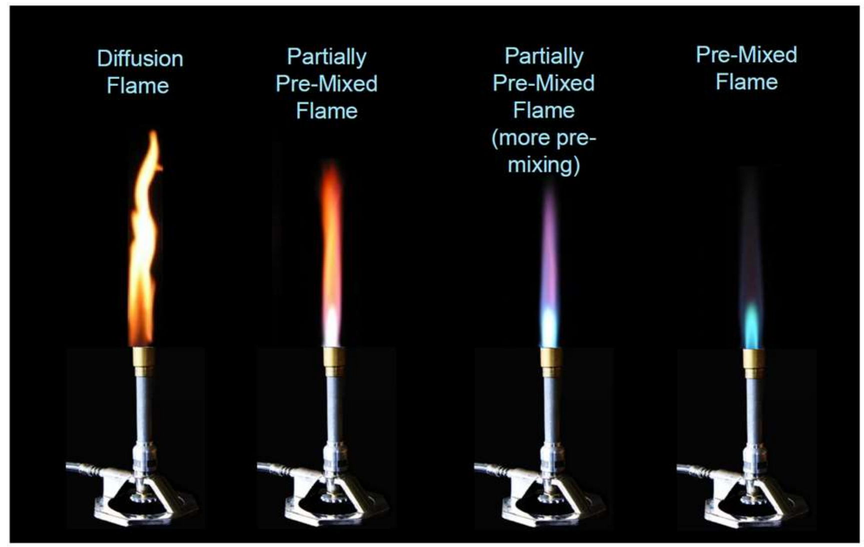
Deflagration Flames

<u>Diffusion Flames</u>

- Lower flame temperatures
- Wider stability limits
- Higher carbon content in flame
- Higher radiation (luminous emissivity)
- Rate of combustion influenced by Fuel/air mixing rate

<u>Pre-Mixed Flames</u>

- Higher flame temperatures
- Narrower stability limits
- Lower carbon content in flame
- Lower radiation (luminous emissivity)
- Combustion influenced by chemical kinetic factors

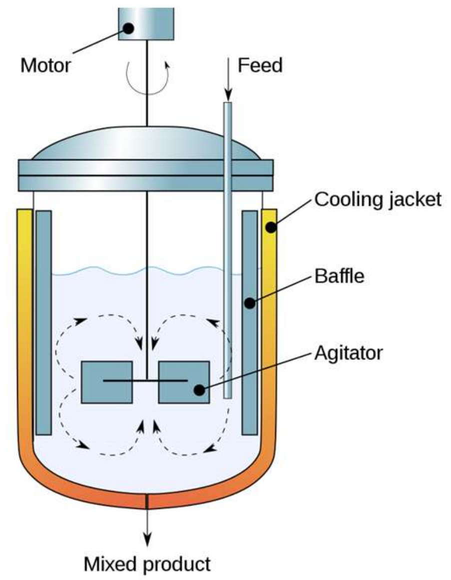


Flame characteristics: Stirred Reactor Theory

The primary zone of a burner and the wake of a bluff body in the afterburner are modeled as a *perfectly stirred reactor* where the reactants and products are mixed infinitely fast. Such an environment produces the maximum energy release (due to infinitely fast chemical reaction rates) for the given volume of the reactor and at a fixed reactor pressure. The theory of perfectly stirred reactor has many applications, among which the stability of premixed turbulent flames in a flow environment.

The perfectly stirred reactor theory is as follows, commonly referred to air loading,

$$\frac{\dot{m}}{p^2 V} = I$$

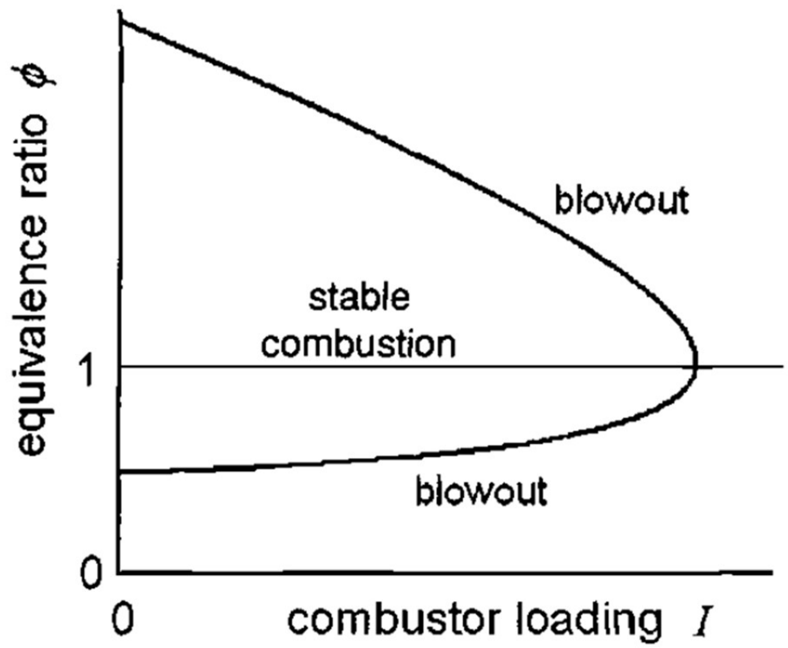


Flame characteristics: Combustor Loading Parameter

In gas turbine practice the air loading concept is applied to the entire combustor when so, it is mostly called the combustor loading parameter (CLP)

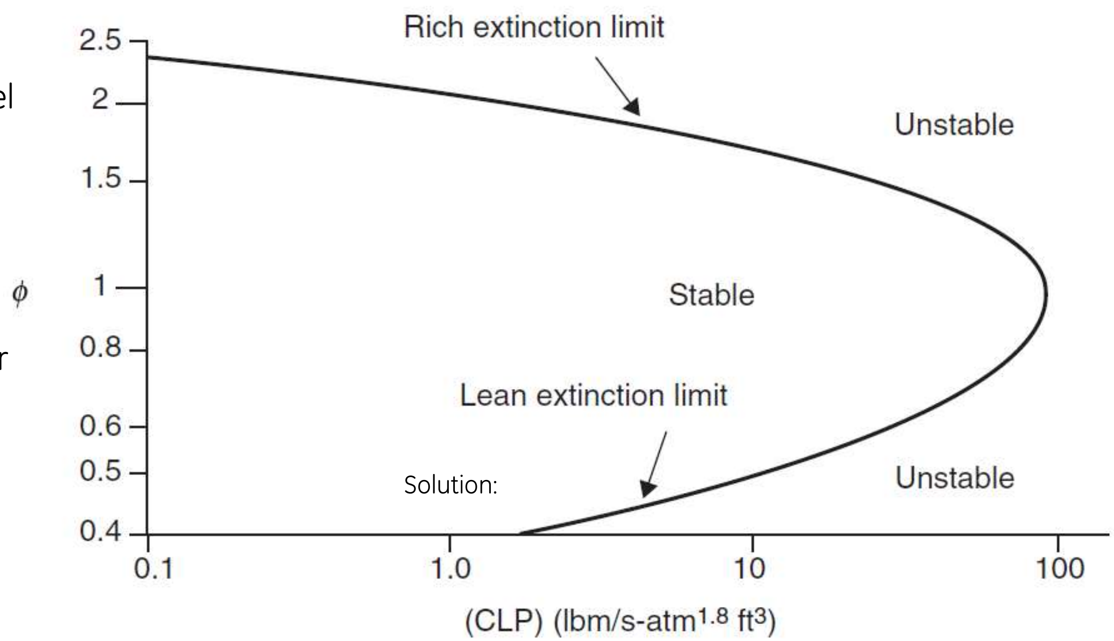
$$CLP = \frac{\dot{m}}{p^n V}$$

The exponent n in the equation has been found empirically to be closer to 1.75-1.8 than 2. Because the entire combustor does not satisfy the assumptions necessary to model the primary zone as a perfectly stirred reactor.



Example

Mass flow rate (combined fuel and air) into a reactor of volume 2.35 ft3 is 100 lbm/s. The equivalence ratio is 0.6. Use the combustor loading parameter (CLP) and the combustor stability loop to establish the range of reactor pressure that leads to a stable combustion.



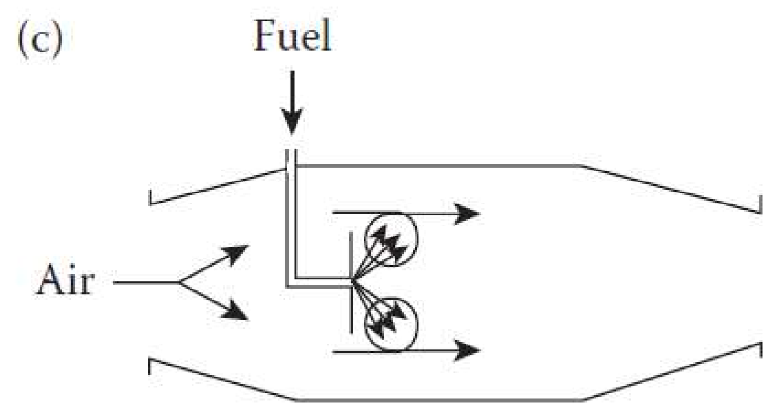
Solution:

Basic Design Criteria

Stable flame

Challenge?

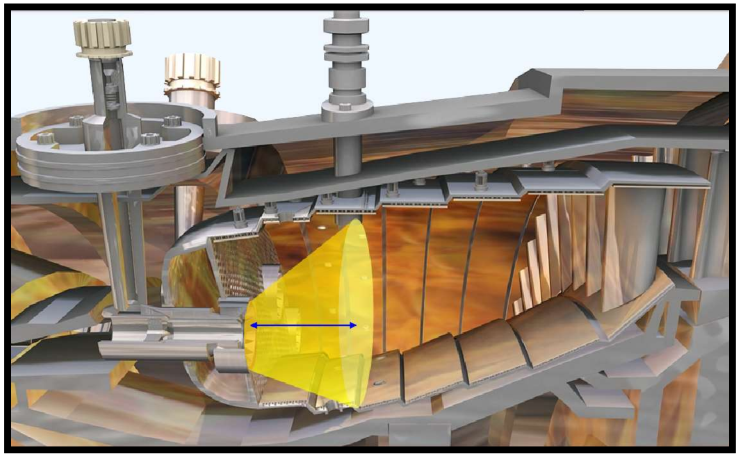
- Not enough temperature



Combustion Intensity

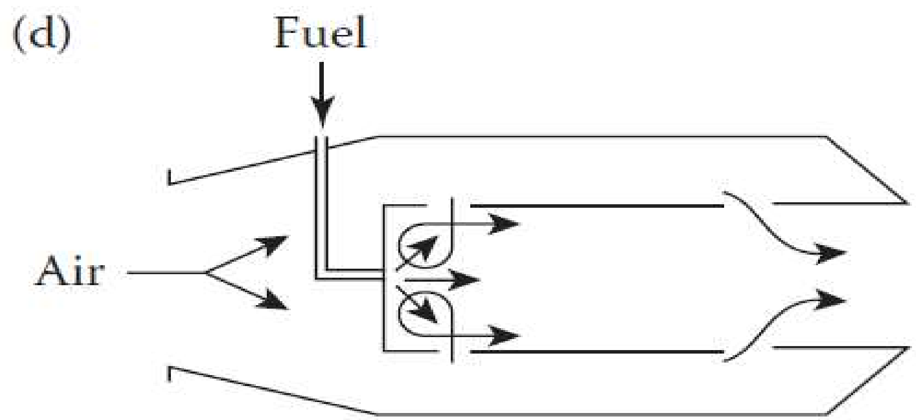
The heat released by a combustion chamber is dependent on the volume of the combustion area. So, the combustion area must be increased. Therefore, the average velocities increasing in the combustor will make efficient burning more and more difficult.

Generating turbulent flow ensures the proper fuel mixing area, obtaining a better distribution of temperature.

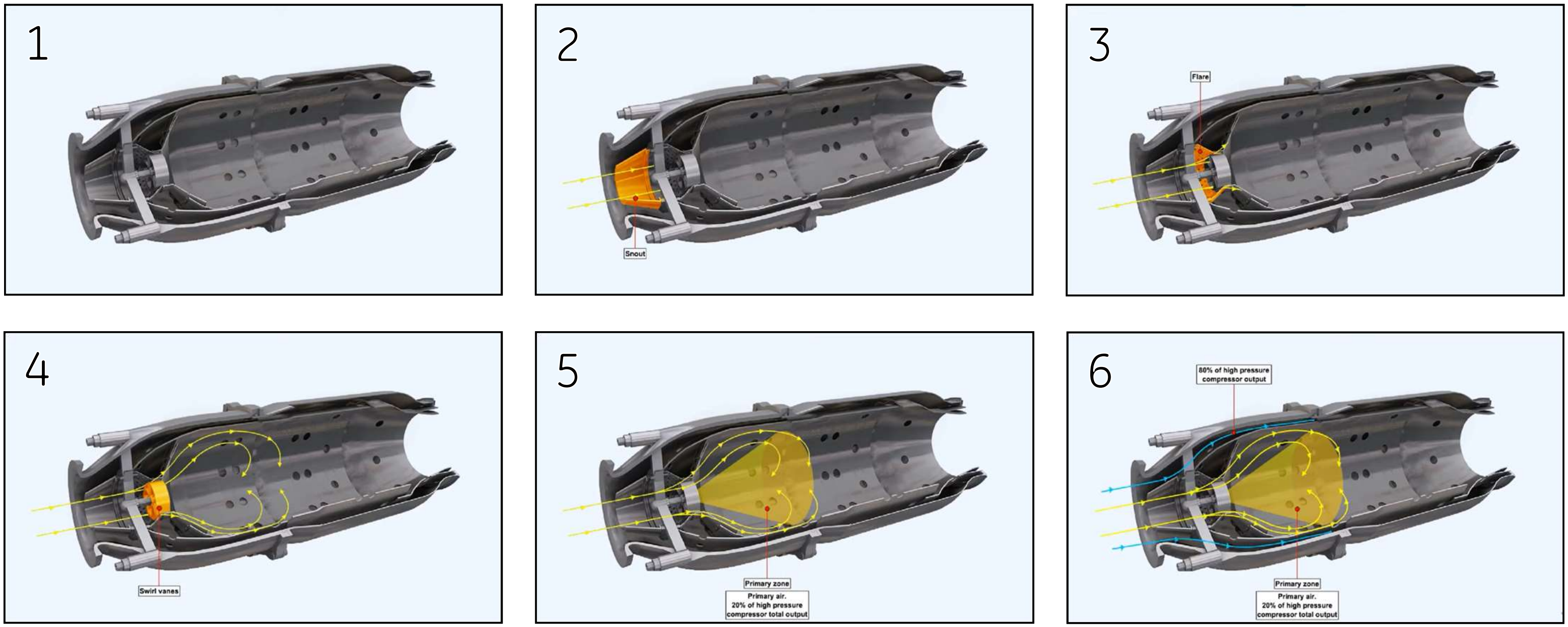


Basic Design Criteria

CURRENT DESIGN

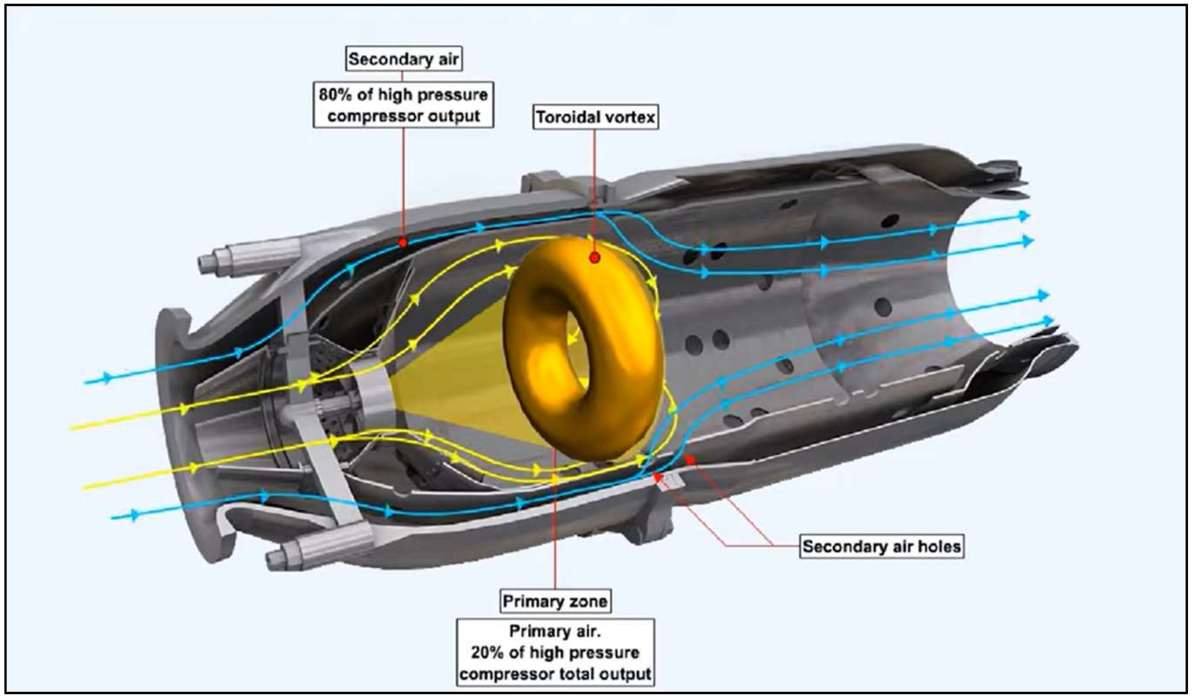


How it Works?



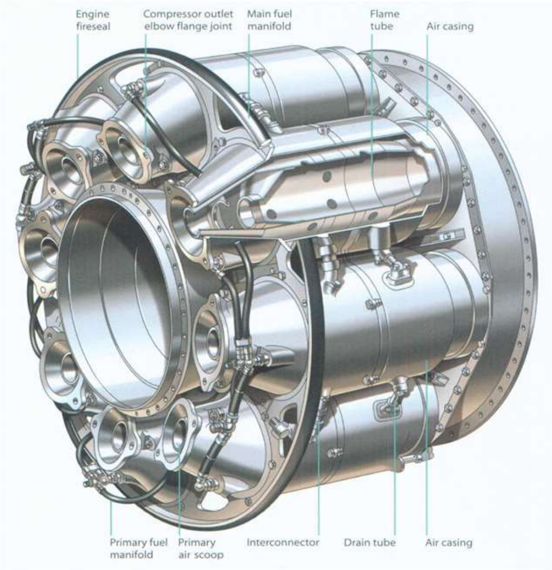
1
2
Snout
3
Flare
4
Swirl vanes
5
Primary zone
Primary air.
20% of high pressure
compressor total output
6
80% of high pressure
compressor output
Primary zone
Primary air.
20% of high pressure
compressor total output

How it Works?

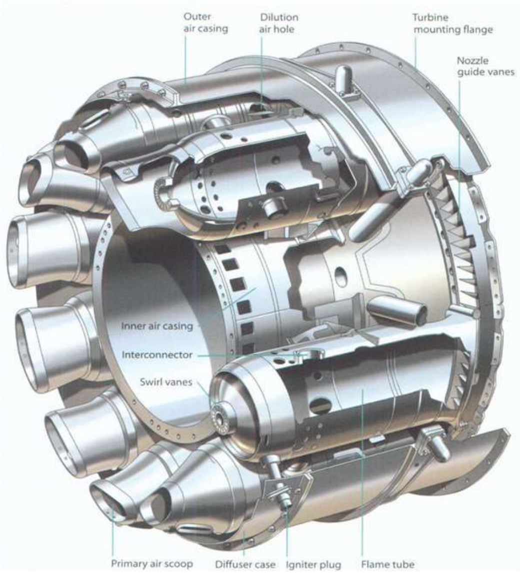

Secondary air
80% of high pressure compressor output
Toroidal vortex
Secondary air holes
Primary zone
Primary air.
20% of high pressure compressor total output

Combustors Types

CANNED

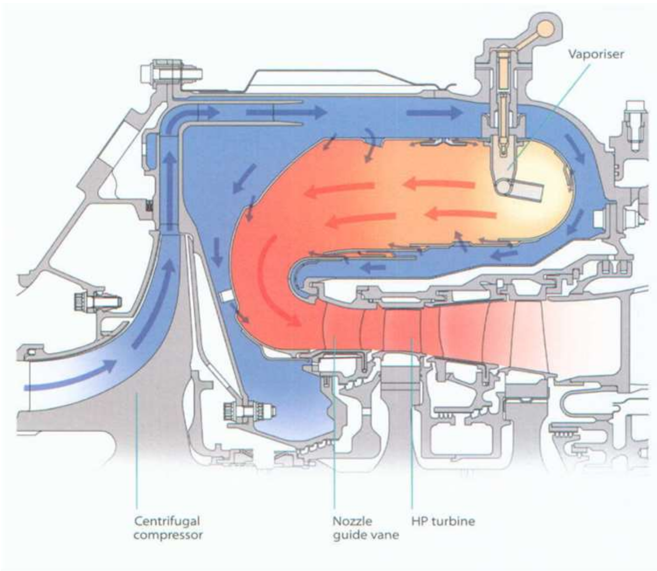


CAN ANNULAR

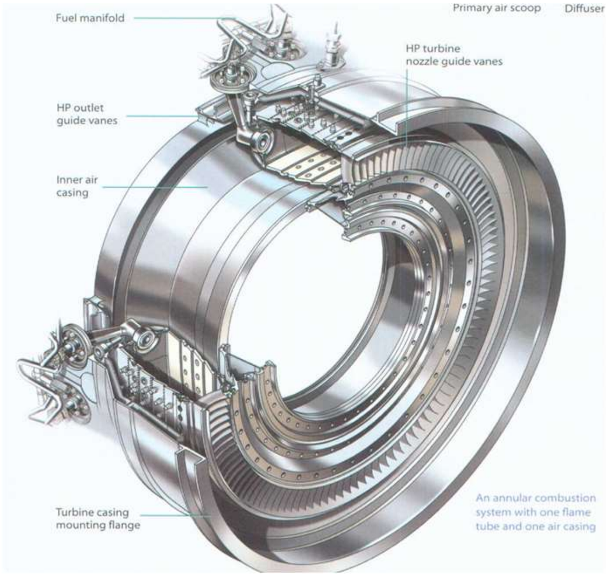


Combustor Types

REVERSED FLOW



ANNULAR



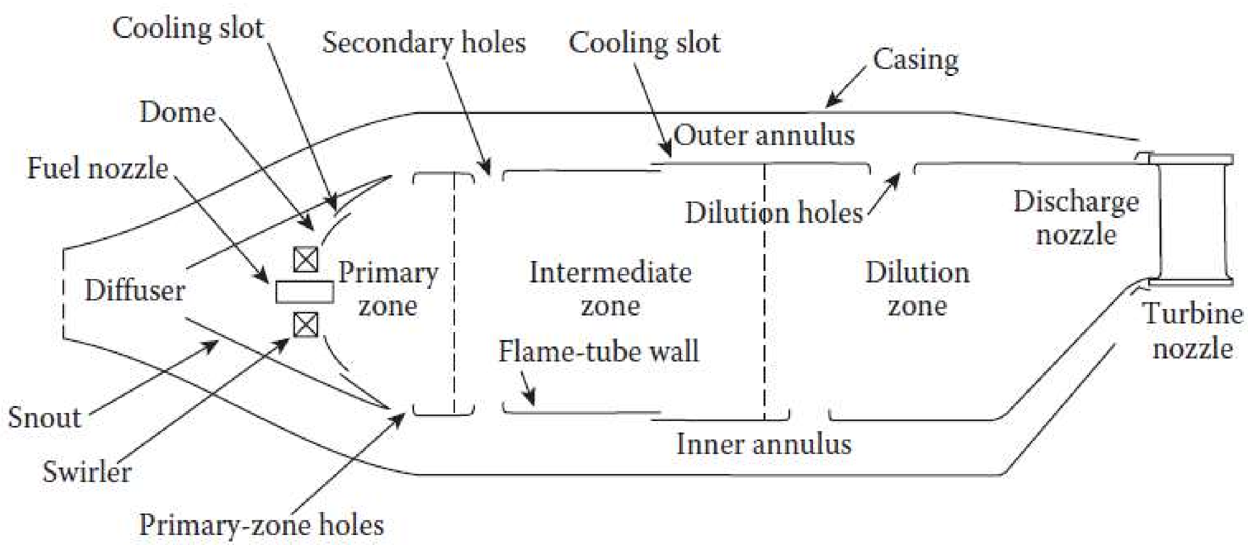
Combustor Types (Comparison)

100%

75%

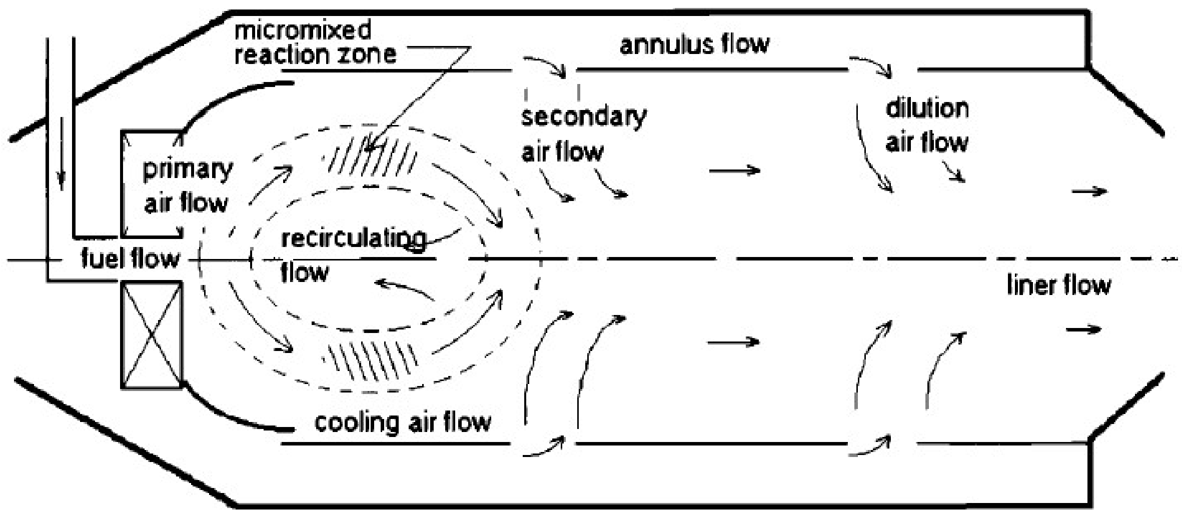
Anatomy

Main components of a conventional combustor



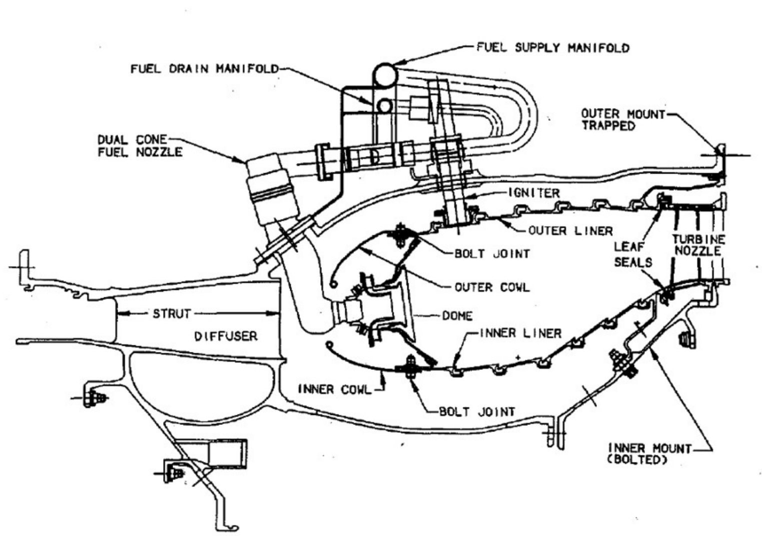
Anatomy

Flow Patterns of a conventional combustor

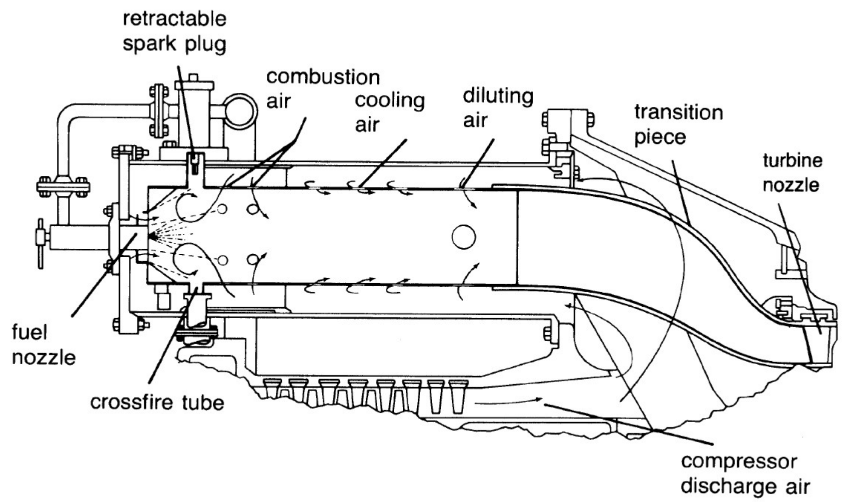


Examples

AERO COMBUSTOR (CF6-80C)



TYPICAL INDUSTRIAL COMBUSTOR

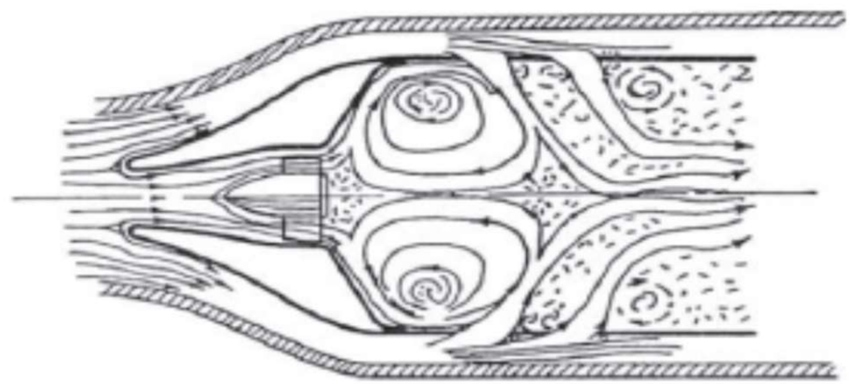


Primary Zone

Anchors the flame

Provides sufficient time, mixing, temperature for “complete” oxidation of fuel

Equivalence ratio near 0.8 - 1



Intermediate Zone

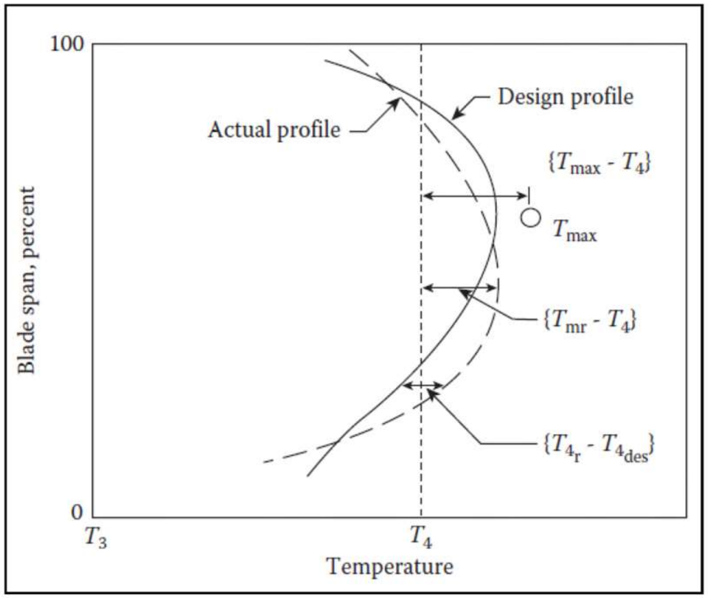
The main functions in this area are:

- **Low altitude** operation (higher pressures in combustor)
  - Recover dissociation losses (primarily $CO \rightarrow CO_2$) and Soot Oxidation
  - Complete burning of anything left over from primary due to poor mixing
- **High altitude** operation (lower pressures in combustor)
  - Low pressure implies slower rate of reaction in primary zone
  - Serves basically as an extension of primary zone (increased tres)
  - L/D ~ 0.7

Dilution zone

The relevance of this area lies in its the most important functions are:

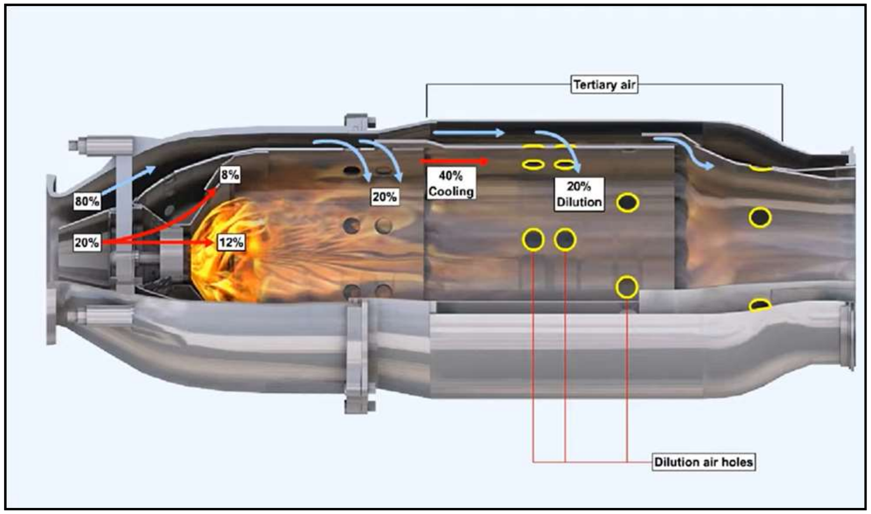
- Dilute the combustion gases with a substantial amount of air.
- Provide a uniform temperature profile
- Provide a suitable radial temperature distribution that does not affect the life of the individual components of the turbine.



$$\text{Turbine profile factor} = \frac{(T_{4.r} - T_{4.des})_{max}}{T_4 - T_3}$$

Combustion Characteristics

Each combustor is designed with the amount of air that is adapted according to the operation.



Combustor Design Process

The design of the main burner can be approached in the same manner as other components of the aircraft engine, this philosophy and approach may be summarized in three steps:

1. From the preliminary design requirements of the engine, identify mass flow rates, pressures and temperatures that the main burner must produce.
2. Use aerodynamics, chemical reactor theory, chemical equilibrium thermodynamics and chemical kinetics to design the processes the working substance must follow to meet the requirements of Step 1.
3. Layout the geometry of walls and passages in such a way as to cause the working substance to follow the processes designed in Step 2.

Combustor Design Process

Identify the constraints

- Overall engine design considerations have already fixed the flow rates of fuel and air that must enter the main burner, the combustion product gases exiting the burner are also by the first stage turbine nozzle entry throat.

Combustor Design Process

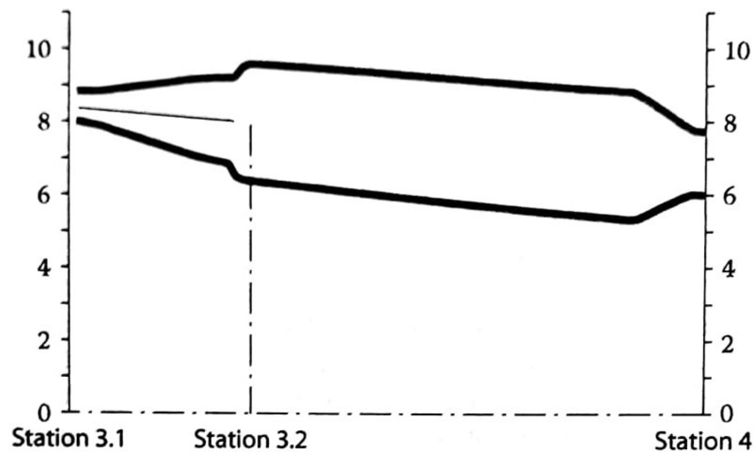
Layout

- The general layout of the main burner will be directly along a line connecting the mean radii of the compressor exit and the turbine inlet.
- Using annular geometry relations, the inner and outer radii and the radial heights can be calculated.

$$r_m = \frac{1}{2}(r_i + r_o)$$

$$H = (r_o - r_i)$$

$$A = \pi(r_o^2 - r_i^2)$$



Note that station 3.2 is the reference condition for main burner calculations

Combustor Design Process

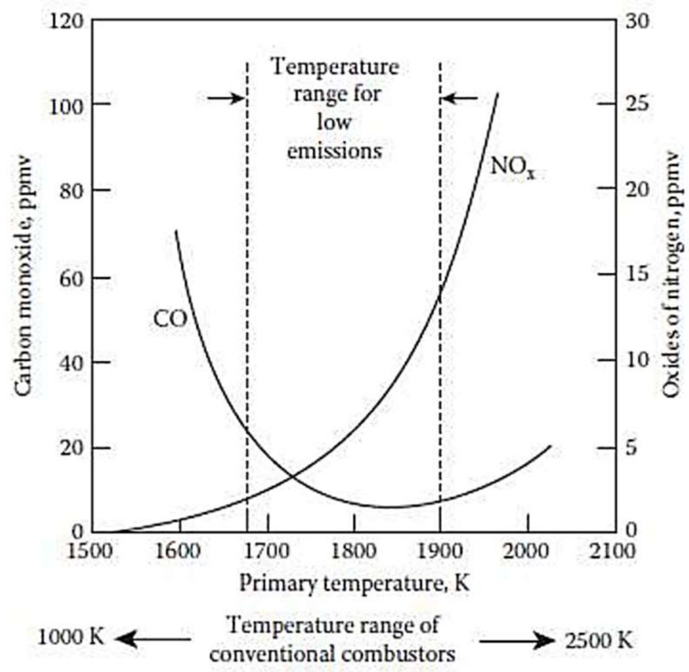
Air partitioning

- The objective of air partitioning is to control the temperatures that the combustion product gases must either achieve, or be limited to, in order to fulfill cycle requirements, ensure the integrity of liner and casing materials, and protect the first-stage stators (nozzles) in the high-pressure turbine immediately downstream of the combustor. By design, only the primary and secondary airflows participate in the combustion process. The liner cooling and dilution airflows are provided only for thermal protection of the liner and of the high-pressure turbine, respectively.

Combustor Design Process

Air partitioning challenges

- If liner gas temperatures are too low, flame stability will be adversely affected, and combustion will not be complete at burner exit. Not only will combustion efficiency be reduced, but also the products of incomplete combustion will include excessive amounts of the pollutants carbon monoxide (CO) and unburned hydrocarbons (UHC).
- If liner gas temperatures are too high, the structural integrity of the liner will be jeopardized, and the emitted air pollutants nitrogen oxide (NO) and nitrogen dioxide (NO2), collectively termed "NOx", may exceed prescribed limits.



Combustor Design Process

Primary Zone

- The governing design principle for partitioning air for the primary zone is the Bragg Criterion, which states: *The maximum overall combustor efficiency occurs when the primary zone is operating at incipient blowout.*

Because today's gas turbines today can operate at much higher temperature levels than in the past, lower values of the local combustion efficiency at blowout ε_{BO} are possible

- The approximate energy equation to be used to guide the design choices is as follows:

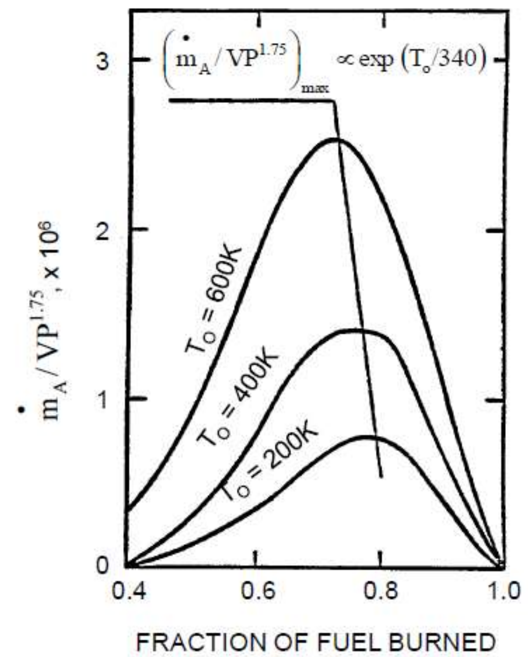
$$T = T_{in} + \varepsilon\phi(T_{ad} - T_{in})$$

where ε is the "fraction reacted" or combustion reaction also commonly known as the combustion efficiency and ϕ the equivalence ratio.

Combustor Design Process

Reaction Rate and Combustion efficiency

- From inspection of the graph, it can be seen that there is an optimal value of ε between 0 and 1 for which RRf is a maximum. As one might guess, a design goal for afterburners and combustor primary zones is to achieve as nearly as possible this maximum value of RRf at some location in the combustion device.



Combustor Design Process

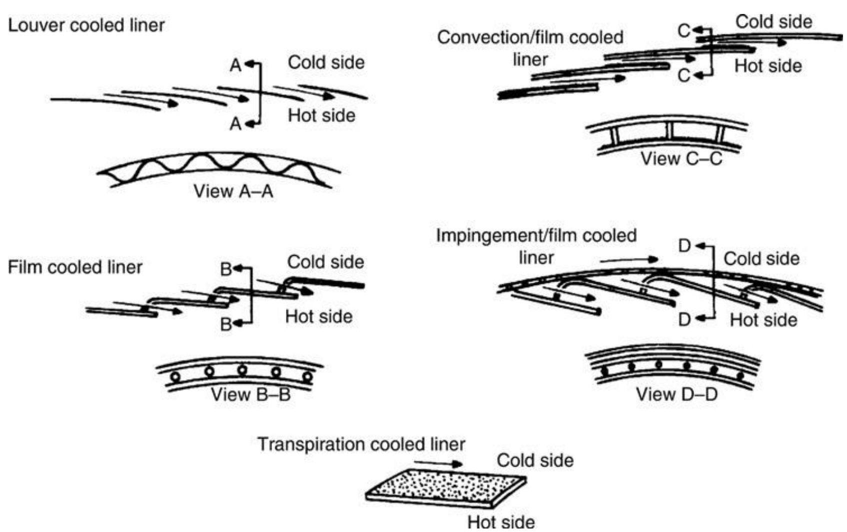
Wall Cooling

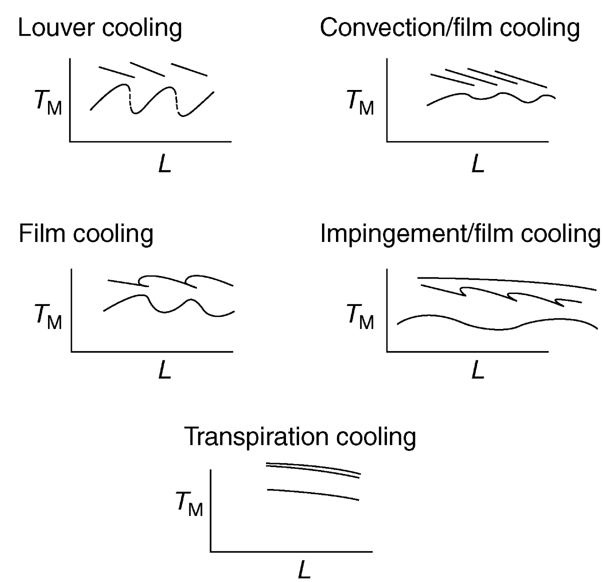
The need

- Combustion takes place inside a combustor liner. It consists of a dome in the primary zone, primary air holes for the primary (radial) jets, air holes for the intermediate zone (radial jets), and the cooling holes/slots in the dilution zone. The combustion temperature may exceed 2500 K (in the primary zone), whereas the liner temperature needs to be kept at ~1200 K.
- The conventional cooling techniques used in gas turbine engines are
  - Convective cooling
  - Impingement cooling
  - Film cooling
  - Transpiration cooling
  - Radiation cooling
  - Some combination of the above.

Combustion Design Process

The coolant mass flow requirement is diminished with cooling effectiveness. Simple film cooling, convection/film, impingement/film, and the transpiration cooling are ranked from the highest to the lowest coolant requirement, respectively.





Combustion Design Process

Under steady-state conditions, the rate of heat transfer into a wall element must be balanced by the rate of heat transfer out.

Therefore,

$$(R_1 + C_1 + K)\Delta A_{W_1} = (R_2 + C_2)\Delta A_{W_2} = K_{1-2}\Delta A_{W_1}$$

Assuming:

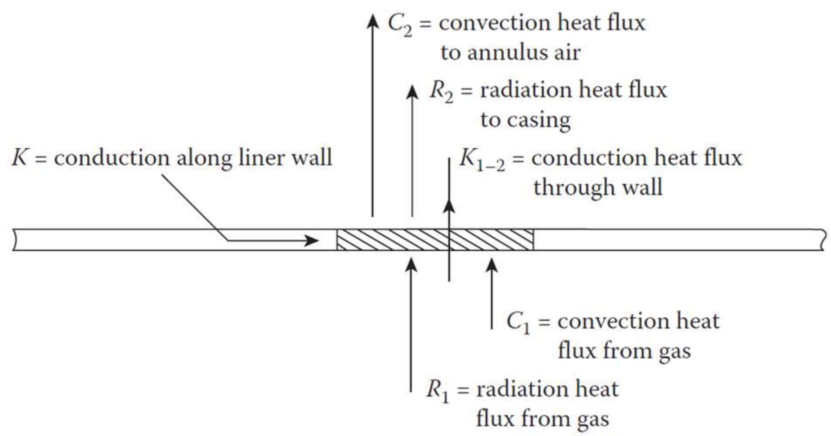
- K is negligible compared to R1, R2, C1 and C2
- Liner wall is usually so thin $\therefore \Delta A_{W_1} = \Delta A_{W_2}$

$$R_1 + C_1 = R_2 + C_2 = K_{1-2}$$

where

$$K_{1-2} = \frac{k_w}{t_w}\left(T_{W_1} - T_{W_2}\right)$$

$k_w = thermal\ Conductivity$
$t_w = liner\ wall\ thickness$

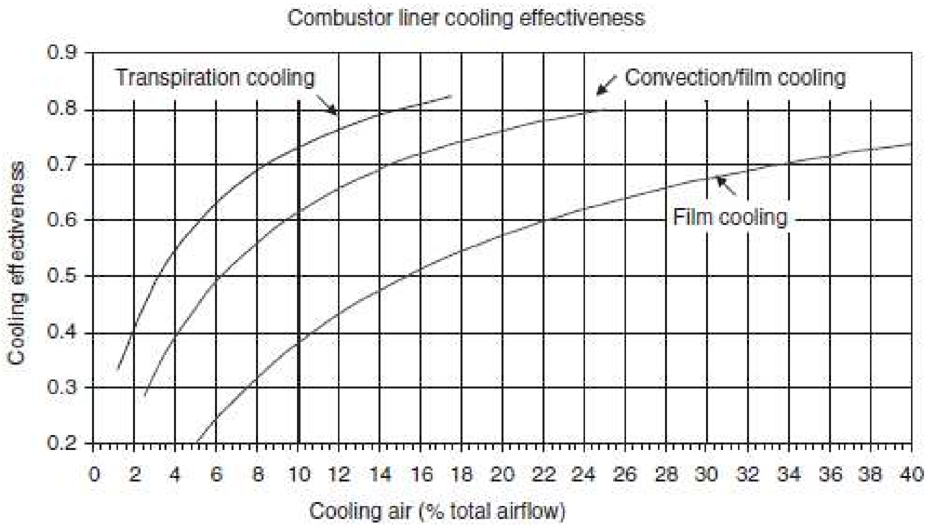


Combustor Design Process

To quantify the coolant mass flow requirement, we define a cooling effectiveness parameter Φ according to

$$\Phi = \frac{T_g - T_w}{T_g - T_c}$$

$$T_w = Liner\ wall\ temperature$$
$$T_g = Combustion\ gases\ temperature$$
$$T_c = coolant\ temperature$$



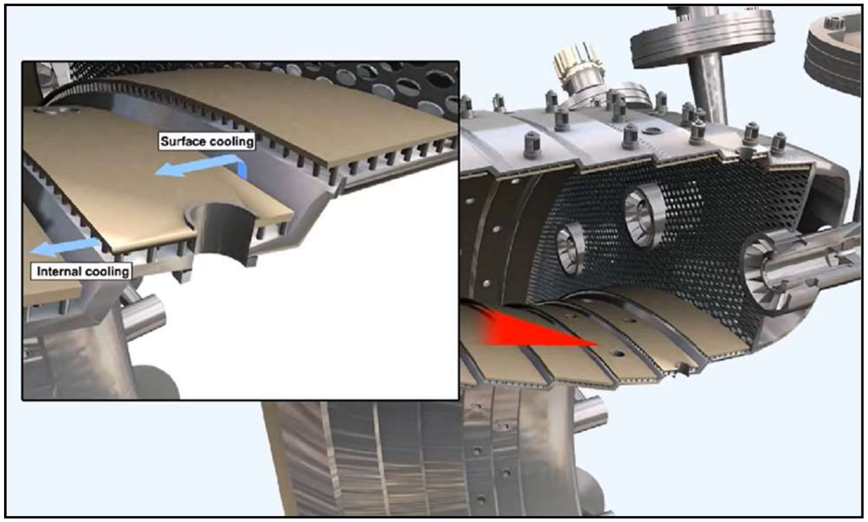
Combustor Design Process

Film Cooling

$$\mu_C = \frac{\dot{m}_c}{\dot{m}_D} = \frac{\Phi}{6\,(1-\Phi)}$$

Transpiration/Effusion

$$\mu_C = \frac{\dot{m}_c}{\dot{m}_D} = \frac{\Phi}{25\,(1-\Phi)}$$



Combustor Design Process

Secondary Zone

- For this zone it is desired to have combustion essentially complete by the secondary zone exit. With ε = 1 substituted into the energy balance, the desired equivalence ratio at secondary zone exit ϕ_{sz} determined to be

$$\phi_{sz} = \frac{T_d - T_{in}}{\Delta T_{max}}$$

Combustor Design Process

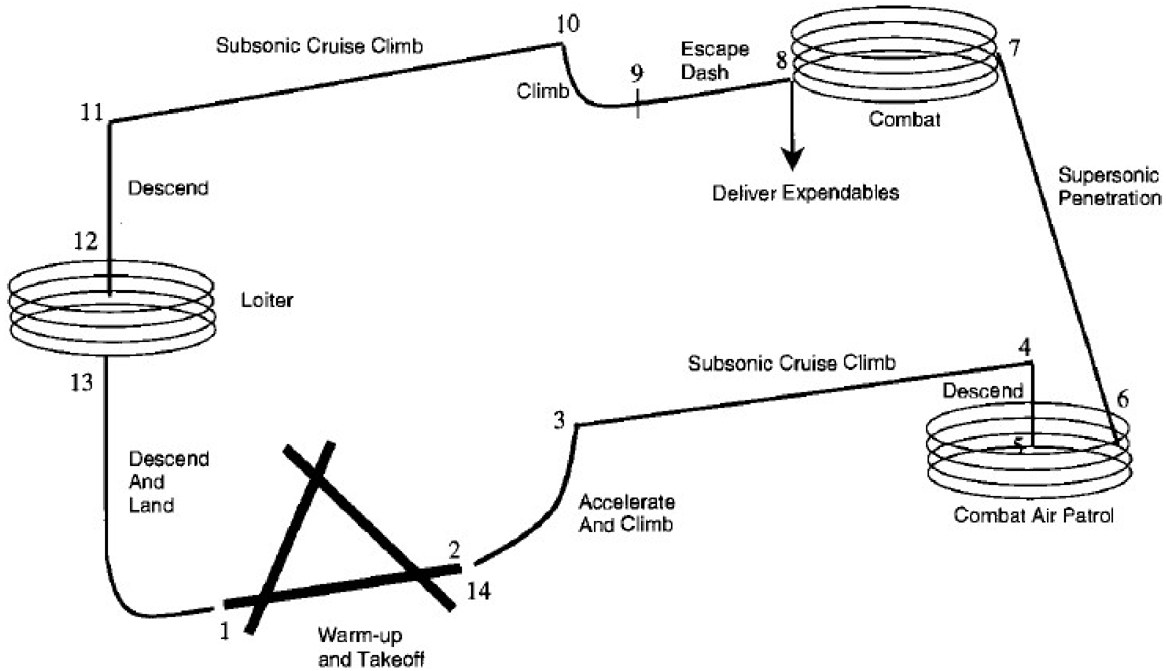
Dilution Zone

- If there is any air left over, it is simply dumped into the dilution zone, however we need to keep in mind continuity and constraints of the design. If there is not enough airflow available to meet all of the demands. First, try to reduce the cooling flow fraction by increasing the cooling effectiveness. Next, adjust the design temperature Tg within the narrow temperature design window, if in the end no dilution air is required, that's all right, because the micromixing in the secondary zone ensures that the "pattern factor" will be zero.

Example

The request for proposal (RFP) mission for the air-to-air fighter (AAF). The flow properties at the identified engine stations for the AAF Engine design point are previously calculated for the engine.

This exercise will just focus on the main burner design.



Example

Mission phases and segments		M_0	Alt, kft	P_{t3}, psia	T_{t3}, °R	T_{t4}, °R	$\dot{m}_{3.1}$, lbm/s	f	$\dot{m}_f$, lbm/s
1–2:	A—Warm-up[a,c]	0.0	2	360.9	1611	3200	63.70	0.02794	1.7795
1–2:	B—Takeoff acceleration[a,b]	0.1	2	361.6	1612	3200	63.81	0.02792	1.7820
1–2:	C—Takeoff rotation[a,b]	0.182	2	363.2	1613	3200	64.10	0.02790	1.7882
2–3:	D—Horizontal acceleration[a,b]	0.441	2	375.0	1625	3200	66.20	0.02771	1.8343
2–3:	E—Climb/acceleration[b]	0.875	23	263.5	1457	2910	48.76	0.02480	1.2090
3–4:	Subsonic cruise climb	0.900	41.6	82.11	1180	2370	16.84	0.01915	0.3224
5–6:	Combat air patrol	0.700	30	84.61	1065	2065	18.59	0.01556	0.2893
6–7:	F—Acceleration[b]	1.090	30	247.2	1475	2948	45.45	0.02524	1.1472
6–7:	G—Supersonic penetration	1.500	30	350.0	1633	3200	61.76	0.02757	1.7024
8–9:	and escape dash[c]								
7–8:	I—1.6M/5g turn[b]	1.600	30	363.0	1650	3200	64.07	0.02729	1.7487
7–8:	J—0.9M/5g turns[b]	0.900	30	199.0	1385	2780	37.67	0.02348	0.8846
7–8:	K—Acceleration[b]	1.195	30	280.9	1532	3056	50.72	0.02641	1.3394
9–10:	Zoom climb[c]	1.326	39	217.5	1525	3054	39.27	0.02648	1.0401
10–11:	Subsonic cruise climb	0.900	47.6	61.22	1177	2406	12.46	0.01983	0.2471
12–13:	Loiter	0.397	10	89.20	980	1734	21.39	0.01131	0.2420
	Maximum dynamic pressure	1.2	0	585.1	1660	3139	104.24	0.02592	2.7017

[a]100°F. [b]Maximum thrust. [c]Military thrust.

AAF Engine Main Burner Operation